

3DES Encryption in Cipher Block Chaining (CBC) Mode

19. Write a C program for encryption in the cipher block chaining (CBC) mode using an algorithm stronger than DES. 3DES is a good candidate. Both of which follow from the definition of CBC.

Which of the two would you choose:

a. For security?

b. For performance?

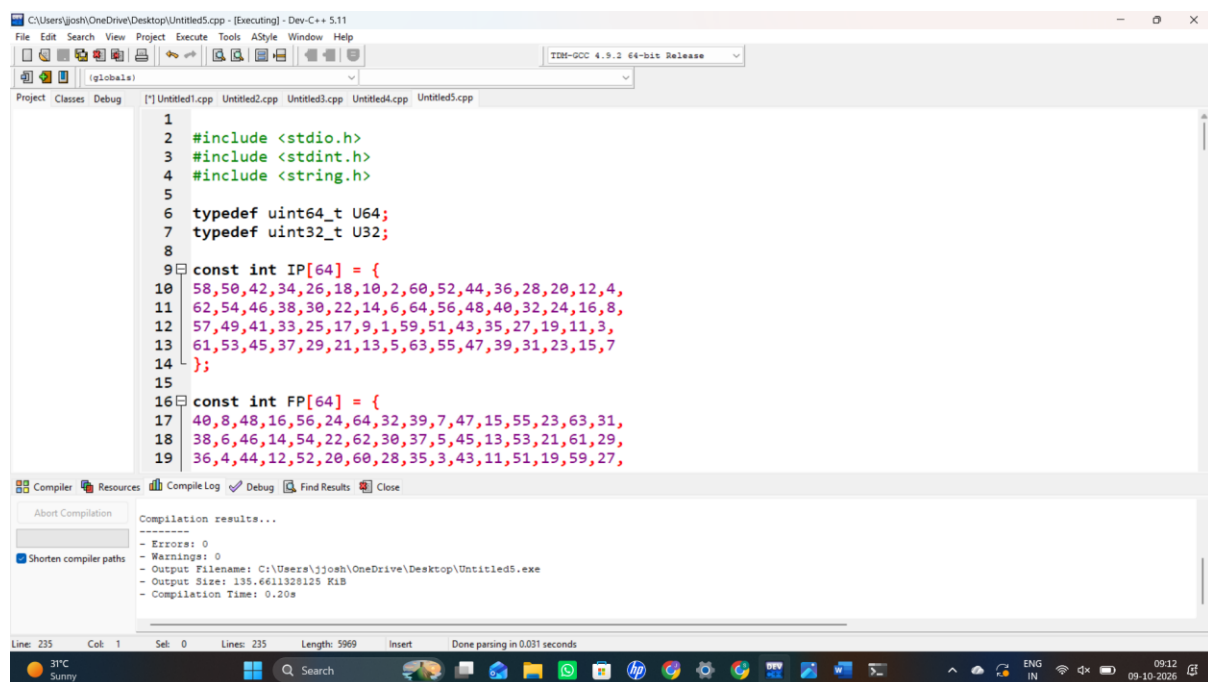
Aim:

To write a C program to encrypt plaintext using Triple DES (3DES) in CBC mode.

Algorithm:

1. Start the program.
2. Read the plaintext, 24-byte key and 8-byte initialization vector (IV).
3. Initialize the 3DES CBC encryption algorithm.
4. Encrypt the plaintext using CBC mode and padding.
5. Display the ciphertext in hexadecimal format.
6. Stop.

Program :

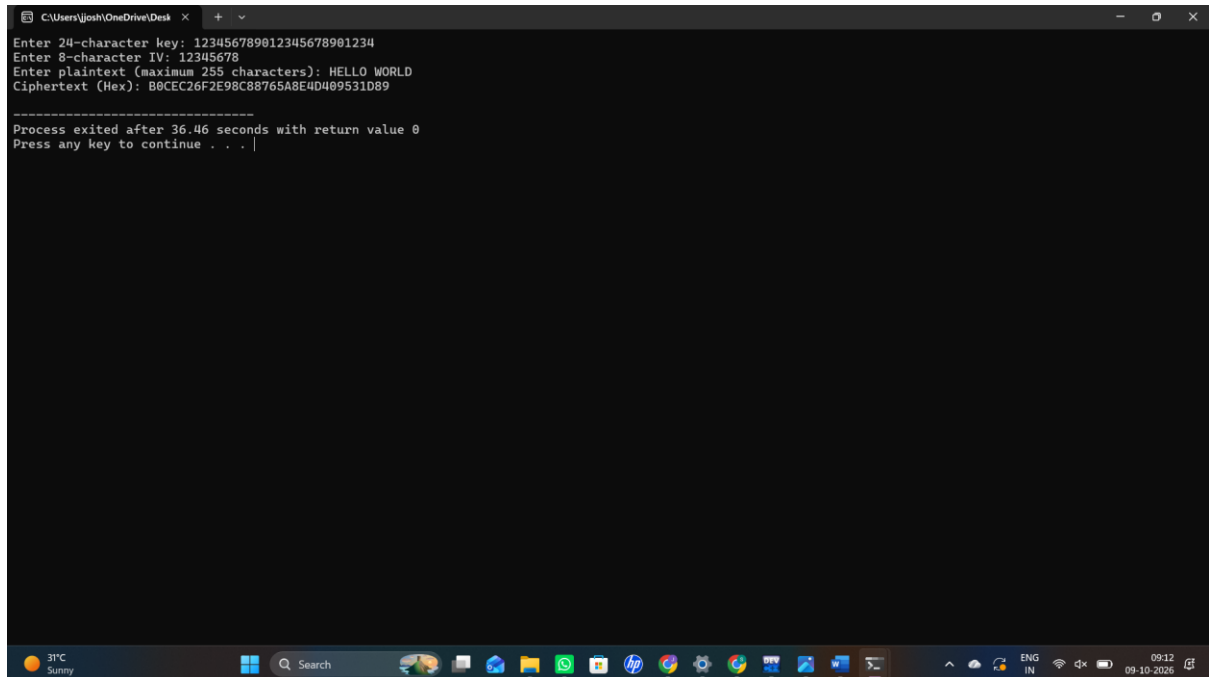


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1
2 #include <stdio.h>
3 #include <stdint.h>
4 #include <string.h>
5
6 typedef uint64_t U64;
7 typedef uint32_t U32;
8
9 const int IP[64] = {
10 58,50,42,34,26,18,10,2,60,52,44,36,28,20,12,4,
11 62,54,46,38,30,22,14,6,64,56,48,40,32,24,16,8,
12 57,49,41,33,25,17,9,1,59,51,43,35,27,19,11,3,
13 61,53,45,37,29,21,13,5,63,55,47,39,31,23,15,7
14 };
15
16 const int FP[64] = {
17 40,8,48,16,56,24,64,32,39,7,47,15,55,23,63,31,
18 38,6,46,14,54,22,62,30,37,5,45,13,53,21,61,29,
19 36,4,44,12,52,20,60,28,35,3,43,11,51,19,59,27,
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```

Compilation results...

Errors: 0
Warnings: 0
Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled5.exe
Output Size: 135.6611328125 KiB
Compilation Time: 0.20s

Input/Output:



```
C:\Users\josh\OneDrive\Desktop > .\3DES.exe
Enter 24-character key: 123456789012345678901234
Enter 8-character IV: 12345678
Enter plaintext (maximum 255 characters): HELLO WORLD
Ciphertext (Hex): B0CEC26F2E98C88765A8E4D409531D89

-----
Process exited after 36.46 seconds with return value 0
Press any key to continue . . .
```

Result:

The program encrypts plaintext using 3DES in CBC mode. It requires OpenSSL development libraries and a build that supports 3DES.

Answers to the questions:

- (a) For security: 3DES is stronger than single DES, although both are legacy algorithms. For a new real-world system, AES is preferred.
- (b) For performance: Single DES is generally faster than 3DES because 3DES performs three DES operations per block. AES is generally the better modern choice.